

DataBase System Concepts

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《数据库系统原理》属于计算机科学与技术学科知识体系中信息管理（IM）知识领域的核心课程。

课程目标：

- 了解数据库技术的现状、发展情况和新技术；
- 掌握数据库系统的基础理论和基本方法；
- 理解数据库管理系统DBMS的核心技术；
- 掌握数据库应用系统设计开发过程；
- 通过课程实验等实践教学环节获得数据库系统使用操作能力和数据库应用系统设计开发能力；
- 培养具有对数据库技术领域复杂工程问题采用科学有效的方法进行研究的能力；
- 对英文技术资料快速阅读和准确理解的能力，以及能够运用英文进行科技交流和沟通的能力。



- Book:

Database System Concepts

- Teaching hours: total 48 hours, including
lecture time
practice time



考核方式:

- 平时作业、实验、期中考试和期末考试
- 平时成绩占总成绩的40%，期末成绩占总成绩的60%
- 总成绩=期末考试60%+期中成绩10%+课后作业10%+实验20%



Reference Book:

数据库系统概念

机械工业出版社

数据库系统概论，王珊著，高等教育出版社

数据库系统原理，李建中，王珊编著，电子工业出版社



Chapter 1 Introduction

Database Structure, Language, Users, Data Model...

Part 1 Relational Languages

Relational Model

SQL

Integrity and Security

Part 2 Database Design

E_R Model

Relational-database Design



Part 3 Application Design And Development

Part 4 Big Data Analytics

Part 5 Data Storage and Indexing

Physical Storage Systems

Data Storage Structures

Indexing



Part 6 Querying Processing And Optimization

Querying Processing

Query Optimization

Part 7 Transaction Management

Transactions

Concurrency control

Recovery System



Part 8 Parallel and Distributed Databases

Part 9 Advanced Topics



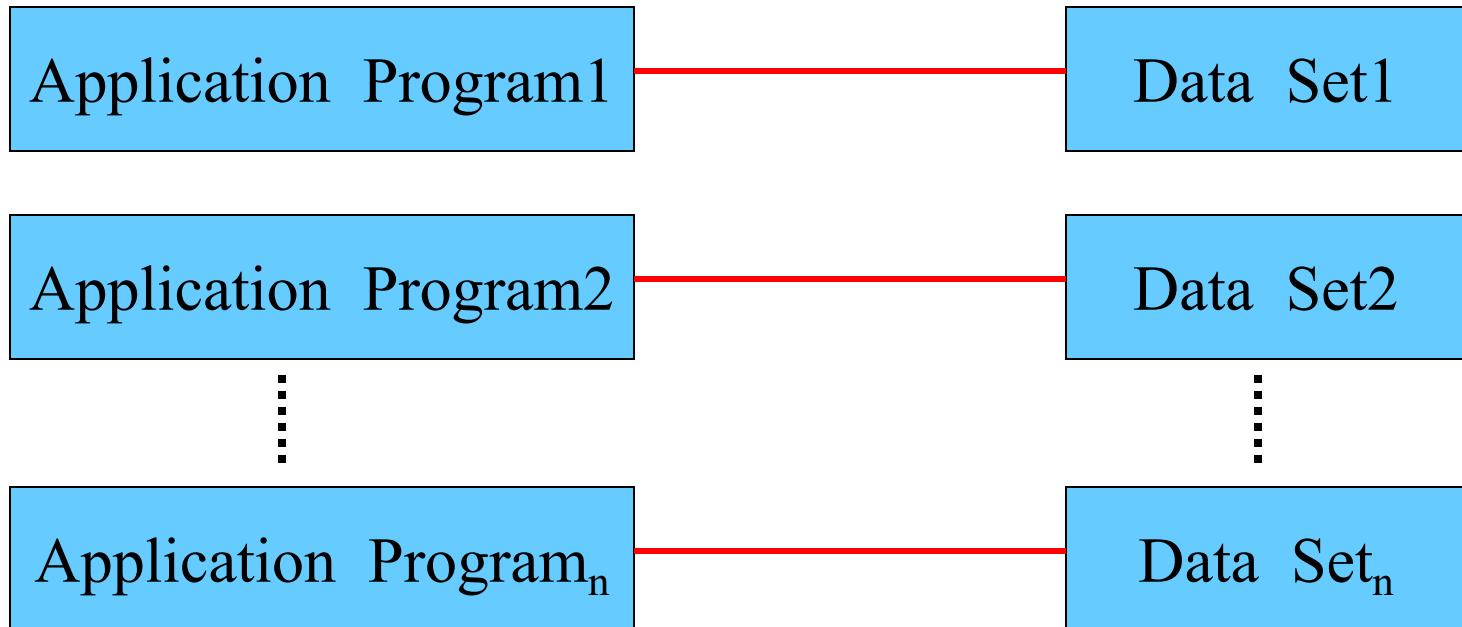
Chapter One Introduction

- ✓ History of Database systems
- ✓ What is Database systems
- ✓ Three levels and Data Abstraction
- ✓ Data Models
- ✓ Database languages
- ✓ Database users
- ✓ Database system structure

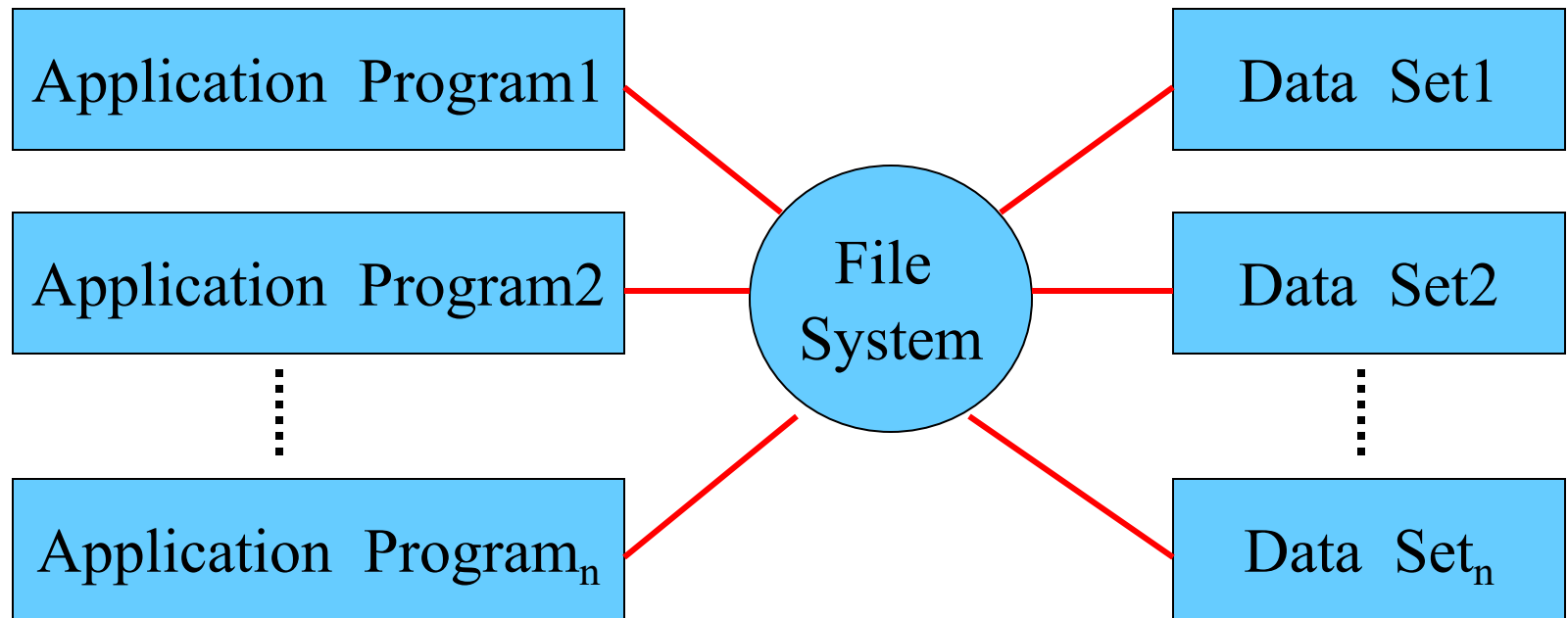


Development of Data Management

Manual Model



File System



Drawbacks of using file systems to store Data:

- **Data redundancy and inconsistency**

Multiple file formats, duplication of information in different files

- **Difficulty in accessing data**

Need to write a new program to carry out each new task

- **Data isolation — multiple files and formats**



- - Integrity problems

- - Integrity constraints (e.g. account balance > 0) become part of program code

- - Hard to add new constraints or change existing ones

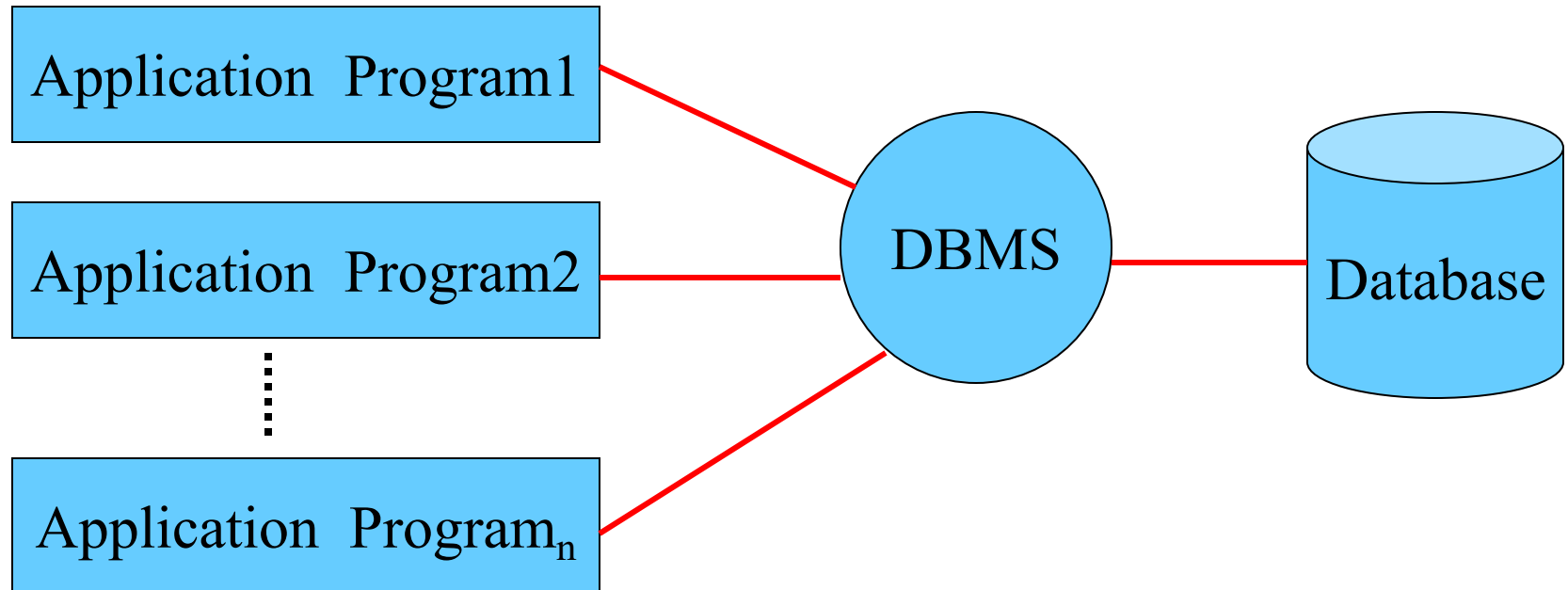
- - Atomicity of updates

- - Concurrent access by multiple users

- - Security problems

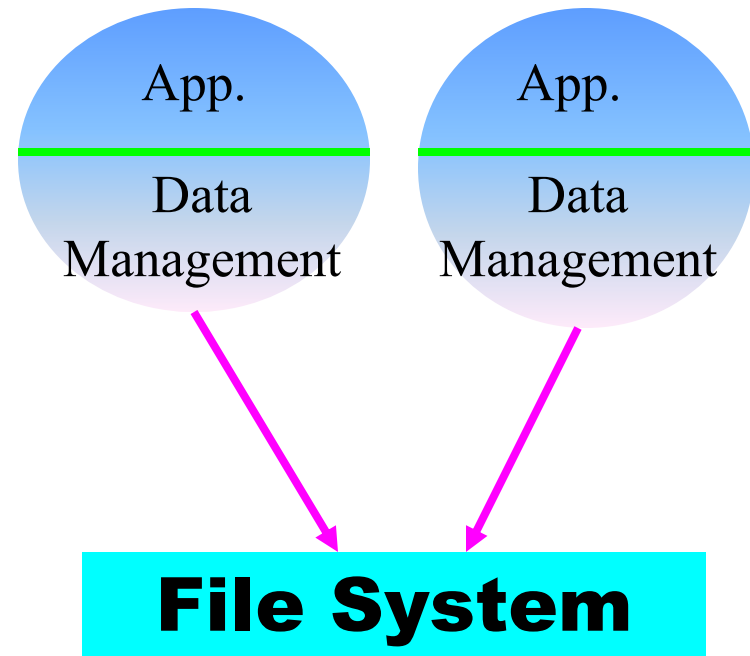
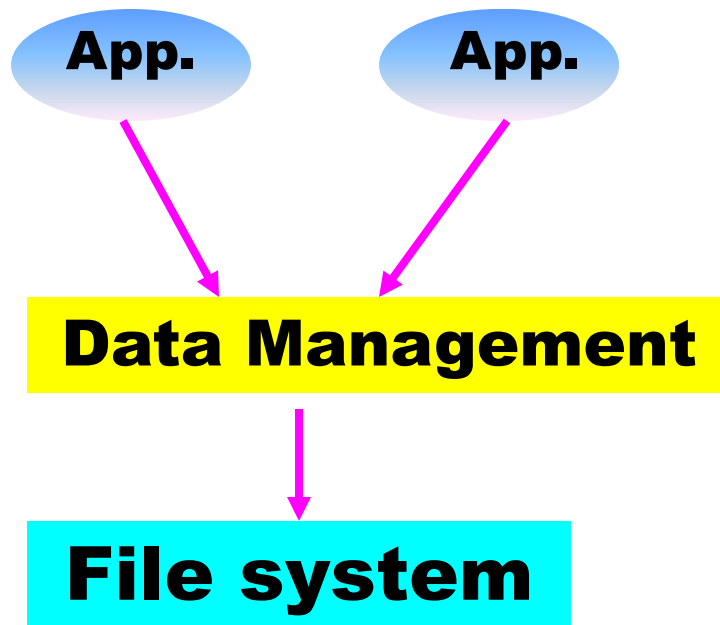


Database System



- Data can be shared;
- Redundancy can be reduced;
- Inconsistency can be avoided(to some extent);
- Transaction support can be provided;
- Integrity can be maintained;
- Security can be enforced;
- ...





- - 1950s and early 1960s:
 - Data processing using magnetic tapes for storage
 - Punched cards for input
 - Late 1960s and 1970s:
 - Hard disks allow direct access to data
 - Network and hierarchical data models in widespread use
 - Ted Codd defines the relational data model
 - Would win the ACM Turing Award for this work
 - IBM Research begins System R prototype
 - UC Berkeley begins Ingres prototype
 - High-performance (for the era) transaction processing

- 1980s:
 - Research relational prototypes evolve into commercial systems
 - SQL becomes industrial standard
 - Parallel and distributed database systems
 - Object-oriented database systems
- 1990s:
 - Large decision support and data-mining applications
 - Large multi-terabyte data warehouses
 - Emergence of Web commerce

- - 2000s
 - XML and XQuery standards
 - Big data storage systems
 - Google BigTable, Yahoo PNuts, Amazon,
 - “NoSQL” systems.
 - Big data analysis: beyond SQL
 - Map reduce and friends
- - 2010s
 - SQL reloaded
 - SQL front end to Map Reduce systems
 - Massively parallel database systems
 - Multi-core main-memory databases
-
-

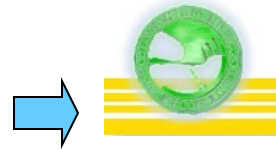
- 1968, IBM's IMS (Information Management system)
hierarchical data model;
- 1970s, Cullinet Software's IDMS
- 1971, CODASYL(Conference on Data System Language)
report network data model;
- 1970, E.F.Codd defined the relational model
- 1990s, object-relational database, WEB database, parallel database
- 2000s, NoSQL
- 2010s, Cloud Database , AI+DB



商业数据库时代: 60年代网状和层次数据库; 70年代关系型数据库成为业界主流; 80年代交易型数据库得到成熟发展, 诞生了一系列的商业公司, **Oracle、DB2、SQLServer**等。数据库以大型机、小型机和商业存储为基础设施, 支撑了商业软件的崛起, 世界开始进入信息时代。

开源数据库时代: 从20世纪90年代到21世纪初, 移动互联网应用爆发, 数据类型不断丰富, NoSQL技术应运而生, 以**MySQL、PostgreSQL、Hadoop、Redis**等为代表, 通过开放源代码的形式, 倡导开源自由精神。

云数据库时代: 近10年, 随着云技术的发展, 数据库在云平台自动化部署、运维, 能够极大地加速用户业务环境上线, 实现生产力的提升。亚马逊的 **Redshift** 和 **Aurora**、阿里云的 **PolarDB**、华为云的 **GaussDB**、腾讯云的 **TDSQL**等。中国数据库开始登上历史舞台。



- **DB:** database

- a collection of interrelated data

- **DBMS:**

- D**ata**B**ase **M**anagement **S**ystem

- **Applications**

- Set of programs to access the data



DBMS(教材上) contains information about a particular enterprise

- Collection of interrelated data
- Set of programs to access the data
- An environment that is both *convenient* and *efficient* to use



- Database Applications:

- Banking: all transactions

- Airlines: reservations, schedules

- Universities: registration, grades

- Sales: customers, products, purchases

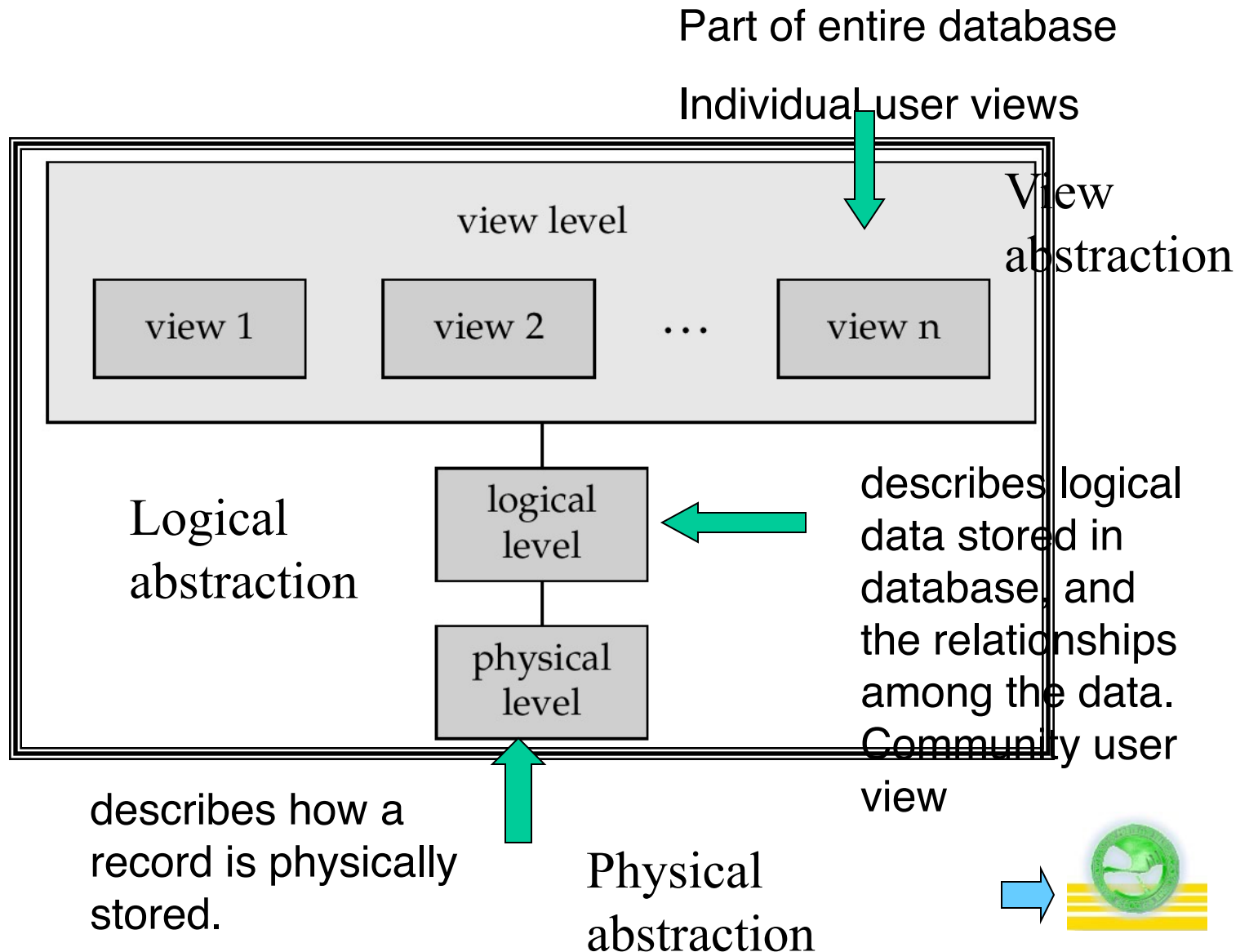
- Manufacturing: production, inventory, orders, supply chain

- Human resources: employee records, salaries, tax deductions

- Databases touch all aspects of our lives



Three-level Architecture



- **Physical level:** describes how a record (e.g., customer) is stored.
- **Logical level:** describes data stored in database, and the relationships among the data.

```
type customer = record  
    customer_id : string;  
    customer_name : string;  
    customer_street : string;  
    customer_city : integer;  
end;
```
- **View level:** Part of the entire database. Application programs hide details of data types. Views can also hide information (such as an employee's salary) for security purposes.

- **View abstraction**
- **Logical abstraction**
- **Physical abstraction**



- **The Physical(internal)/Logical(conceptual) mapping**

defines the correspondence between the logical view and the stored database;

specifies how logical records and fields are represented at the physical level;

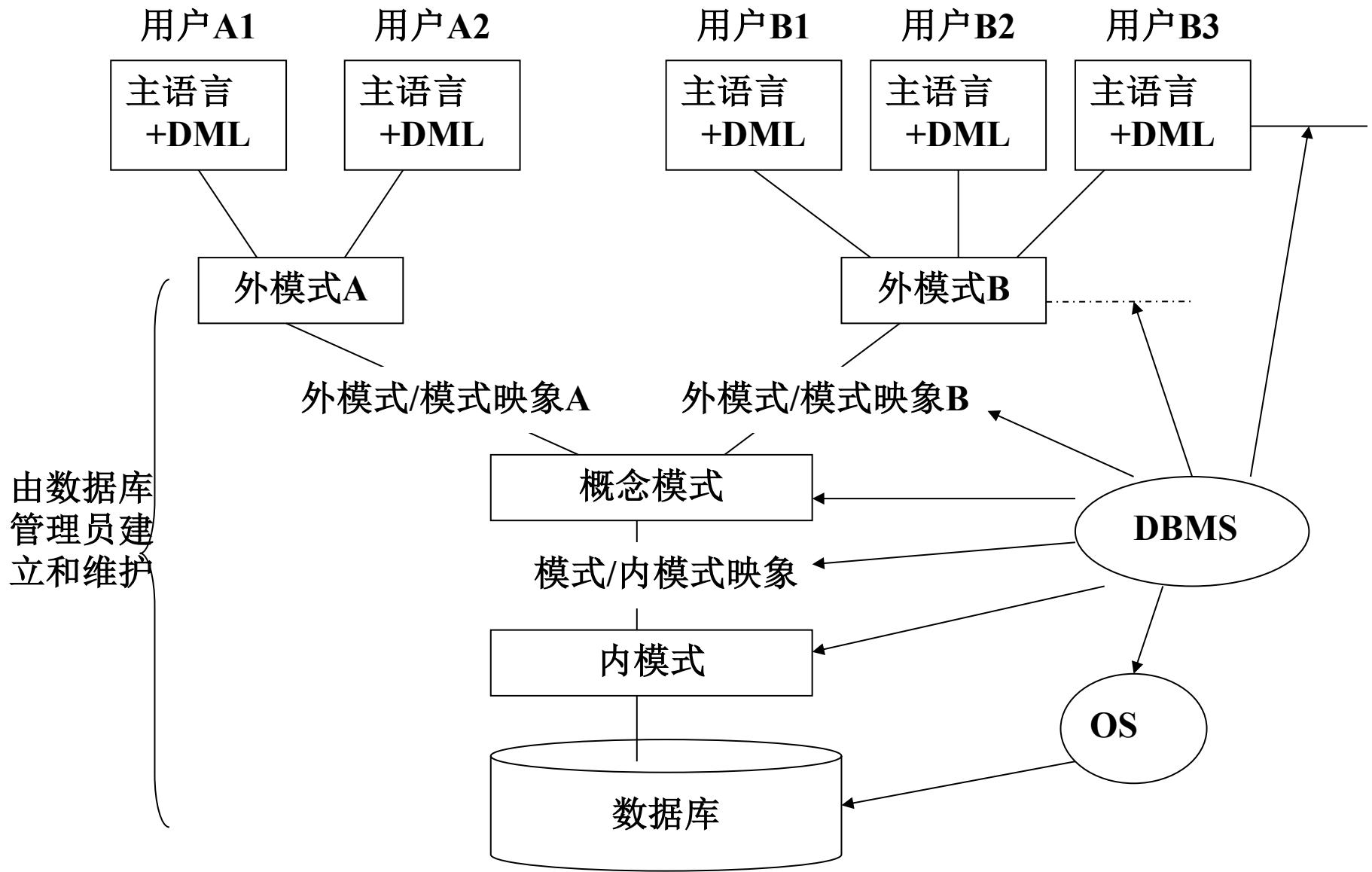
- **The View(external)/Logical(conceptual) mapping**

defines the correspondence between a particular view and the logical view;



- **Physical Data Independence** （物理数据独立性） –
the ability to modify the physical schema without
changing the logical schema or the views
(The Physical(internal)/Logical(conceptual) mapping)
Application programs need not be rewritten if the
physical schema changes.
- **Logical Data Independence** （逻辑数据独立性） –
the ability to modify the logical schema without
changing the logical views
(The View(external)/Logical(conceptual) mapping)





数据库系统的体系结构

- **Schema** (模式) – the description of the structure of the data in a database

- e.g., the database consists of information about a set of customers and accounts and the relationship between them);

- Analogous to type information of a variable in a program;

- **Physical schema:** database design at the physical level;

- **Logical schema:** database design at the logical level;

- **Instance** (实例) – the actual content of the database at a particular point in time

- Analogous to the value of a variable



- A collection of conceptual tools for describing **data, data relationships, data semantics, and consistency constraints.**

- Data Structure;

- Data Operations;

- Constraint Rules.



- Object_based data model:

- Entity-Relational Model (E_R Model);

- Object-oriented Model;

- Record_based data model:

- hierarchical data model;

- network data model;

- relational data model



Relational Data Model

DataBase System Concepts

<i>Customer-id</i>	<i>customer-name</i>	<i>customer-street</i>	<i>customer-city</i>	<i>account-number</i>
192-83-7465	Johnson	Alma	Palo Alto	A-101
019-28-3746	Smith	North	Rye	A-215
192-83-7465	Johnson	Alma	Palo Alto	A-201
321-12-3123	Jones	Main	Harrison	A-217
019-28-3746	Smith	North	Rye	A-201



- **Data Definition Language (DDL):**

- Specification notation for defining the database schema; Eg:

- **Create table** *account* (
 account-number **char(10),**
 balance **integer)**

- Including:

- Database schema

- Data *storage and definition* language



Data Manipulation Language (DML):

- Language for accessing and manipulating the data organized by the appropriate data model;

(query, insert , delete, update)

DML also known as query language

- Two classes of languages

- Procedural – user specifies what data is required and how to get those data ;

- Nonprocedural – user specifies what data is required without specifying how to get those data;

- SQL is the most widely used query language



- Users are differentiated by the way they expect to interact with the system
- •Application programmers – interact with system through DML calls
- •Sophisticated users – form requests in a database query language
- •Specialized users – write specialized database applications that do not fit into the traditional data processing framework
- •Naive users – invoke one of the permanent application programs that have been written previously
 - E.g. people accessing database over the web, bank tellers, clerical staff

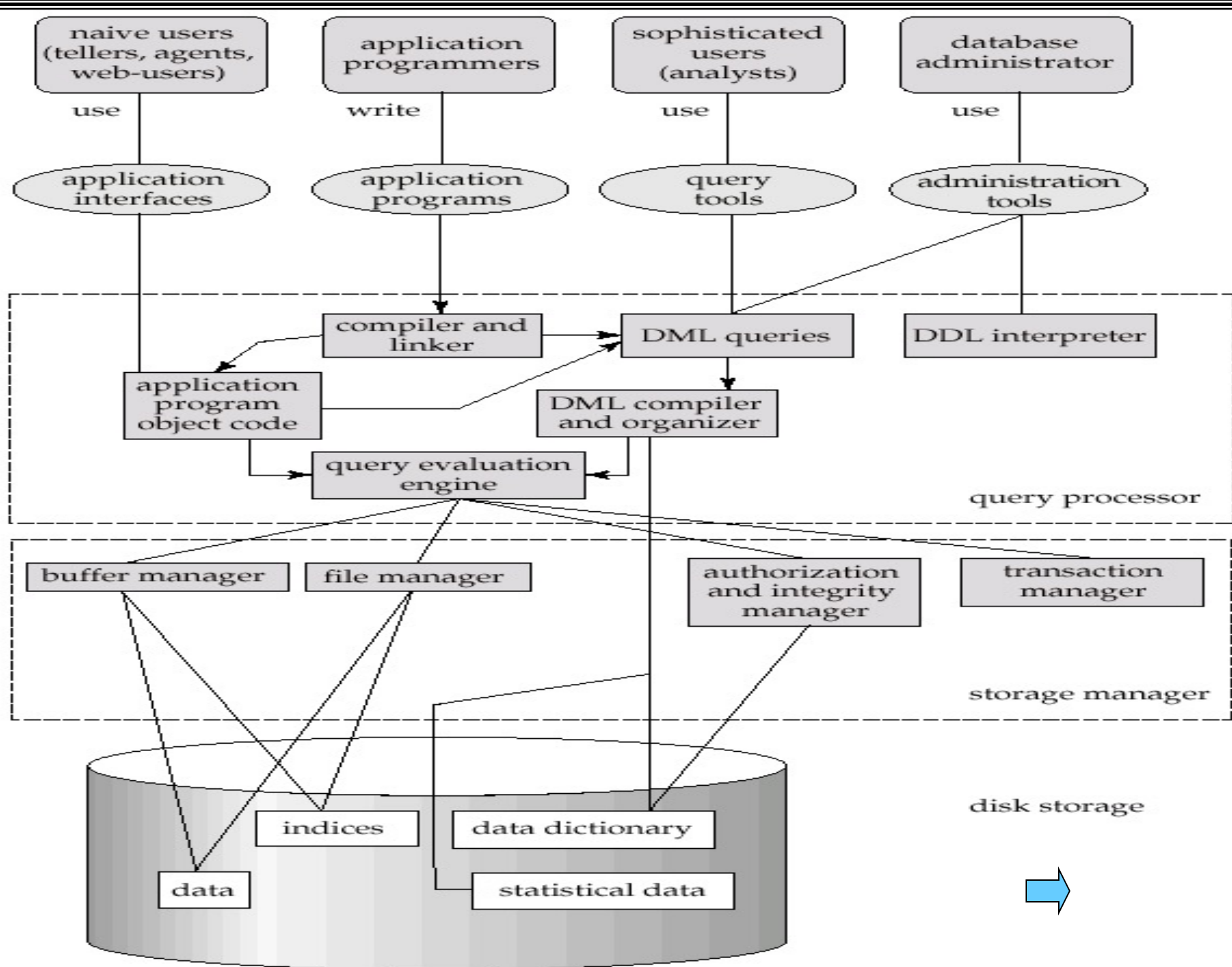


- End Users:
 - naive Users
 - casual users -----Interactive SQL
- Application Programmers-----Procedural SQL, Transaction
- Database analyzer and designer -----Data modeling,
Normalization theory,
- Database Administrators, DBA -----Database maintenance ,
Security , Integrity, Recovery
- Database Management System designer and implementer -----
Implementation technique of above techniques for Special
and New Database Management System



DBA





Storage Management:

- Storage manager is a program module that provides the interface between the low-level data stored in the database and the application programs and queries submitted to the system.
- is responsible for efficient storing, retrieving and updating of data;
- Authorization and integrity manager;
- Transaction manager;
- File manager;
- Buffer manager;
- Data Structure
 - Data files;
 - Data dictionary;
 - Indices;



- **Query Processor:**

- **DDL interpreter;**

- **DML complier;**

- **Query optimization;**

- **Query evaluation engine;**

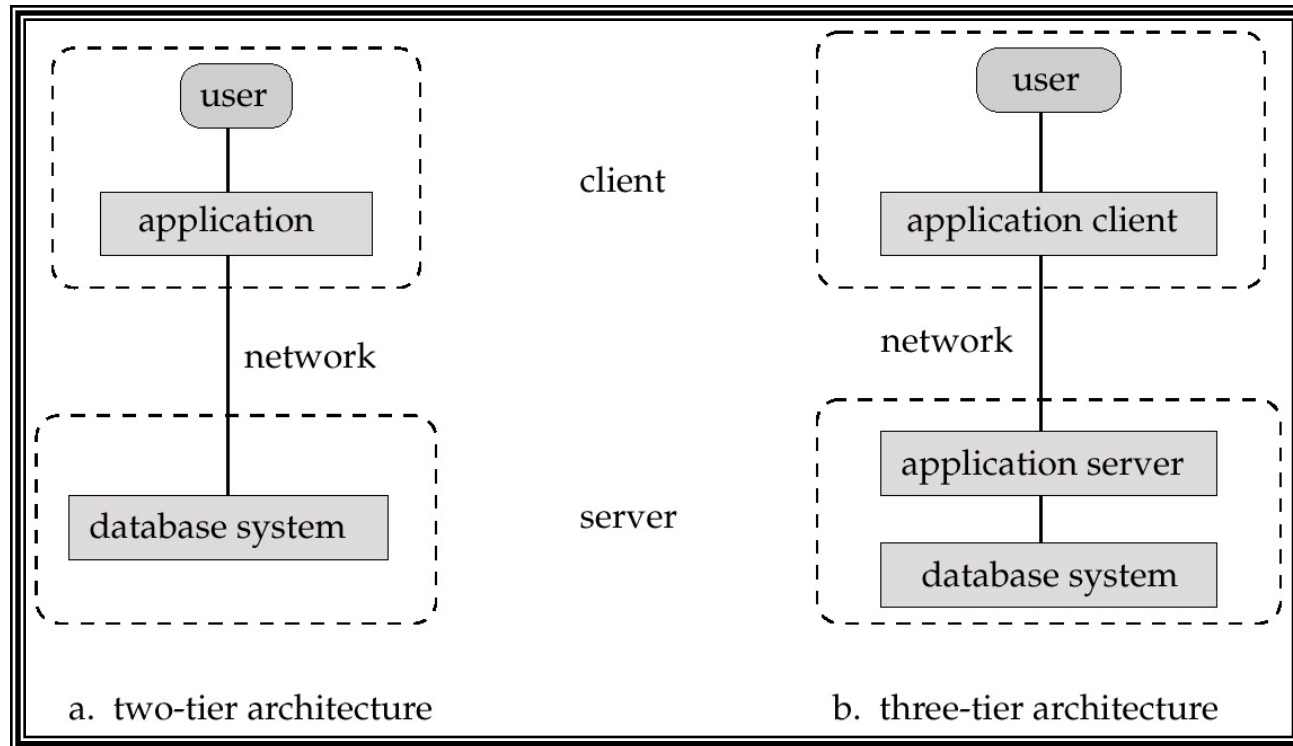


- A *transaction* is a collection of operations that performs a single logical function in a database application
- Transaction-management component ensures that the database remains in a consistent (correct) state.
- Concurrency-control manager controls the interaction among the concurrent transactions, to ensure the consistency of the database.



- **Data Definition and storage management;**
- **Data Manipulation, Data Access;**
- **Data Security and integrity;**
- **Transaction management, Data recovery and concurrency;**
- **Data dictionary;**





- **Two-tier architecture:** E.g. client programs using ODBC/JDBC to communicate with a database
- **Three-tier architecture:** E.g. web-based applications, and applications built using “middleware”



- 网状数据库（IDS 1964）
- 层次数据库（IMS 1968）
- 关系型数据库（市场主流，Oracle、MySQL、SQLServer、DB2、OpenGauss、OceanBase、PolarDB）
- 非关系型数据库 NoSQL
 - ✓ 键值数据库：数据以键值对形式存储，Redis
 - ✓ 文档数据库：以类似JSON的文档格式存储，支持嵌套结构；MongoDB
 - ✓ 列簇式数据库：按列存储，HBase
 - ✓ 图数据库：以图结构存储节点和关系，Neo4j
 - ✓ 时序数据库：针对时间序列数据设计，InfluxDB
 - ✓ 向量数据库：处理向量数据，Milvus



➤ 集中式数据库

数据库集中部署于由一台或多台计算机组成的中心节点，数据整体存储于本地磁盘或外接存储设备。

部署简单、架构成熟；

➤ 分布式数据库

部署于网络中不同计算机上，物理上分散、逻辑上相互关联的数据库，使用者感受不到分布性。



➤ 本地部署

在本地的硬件、网络和其他基础设施上安装的软件服务。

在本地部署模式下，需要先进行硬件采购调试、数据库安装部署等工作，才能就绪和上线数据库应用，企业需要配备专业的数据库管理员(DBA)或通过专业的服务厂商进行

数据库部署和维护服务。

➤ 云部署

数据库在云平台自动化部署、运维。

云数据库可以共享云基础架构，加速数据库的供给和上线，加速用户业务环境上线，简化运维和服务管理。



- 信创

- 信息技术应用创新

- 2016年3月4日，国内24家专业从事软硬件关键技术研究及应用的单位，发起成立非营利性社会组织“信息技术应用创新工作委员会”（信创工委）
 - 随后，全国各地相继成立信创产业联盟，共同催生了庞大的信息技术应用创新产业，被称为“信创产业”

- 信创目的

- 实现我国信息技术产业的完全自主可控，提升信息基础设施安全

信创产业生态



大类	小类	主要企业
基础硬件	芯片	龙芯、飞腾、兆芯、银鹏、海光、申威、景嘉微(GPU)
	存储	长江存储、国科微、兆易创新、北京君正、紫光存储、同有科技、合肥长鑫
	整机(服务器、PC)	华为、浪潮、新华三、中兴、曙光、长城、同方、联想、方正、山西百信、宝德
	固件(BIOS和BMC芯片)	昆仑固件、百敖软件、中国长城、uboot、卓易信息
	外设(打印机、扫描仪)	奔图、长城、紫光、立思辰、纳思达
基础软件	操作系统	华为鸿蒙、麒麟系统、信安科技、深之度、统信软件、普华操作系统、中科方德、中兴新支点、凝思科技、红旗软件、一铭软件, 同源华安、万里红、阿里龙蜥
	数据库	武汉达梦, 人大金仓, 南大通用, 神舟通用, 蚂蚁金服, 阿里云, 华为Gauss DB, PingCAP, 中兴GodenDB, 瀚高基础软件, 优炫数据, 星环
	中间件	东方通、宝兰德、普元信息、中创中间件、金蝶天燕、华宇软件
应用软件	办公软件	金山办公、永中、中标普华office、启明星辰、福昕软件、数科网维
	流版签	金山办公、永中、福昕软件、数科网维、中安网脉、安证通、金格科技
	浏览器	海泰方圆、红芯浏览器
	邮件	中标软件、盈世信息
安全	安全(软件、硬件、服务)	奇安信、深信服、北信源、卫士通、中孚信息、启明星辰、南洋股份、格尔软件、绿盟科技、三六零、海泰方圆、天融信
集成	集成商	中国软件、浪潮软件、太极股份、华迪集团、华宇软件、东软集团、神州信息
云计算	信创云	移动云、天翼云、阿里云、华为云、腾讯云、金山云、浪潮云

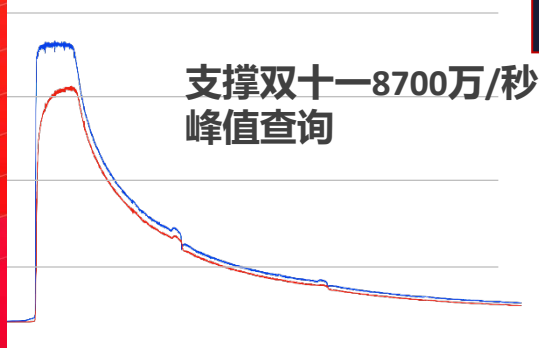
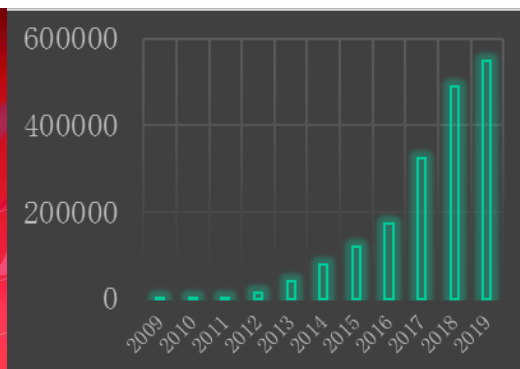
	国外体系
处理器	Intel、AMD、高通、苹果、Nvidia (GPU)
存储	三星、SK海力士、美光、西部数据、铠侠 (东芝)、 EMC
外设	富士、惠普、佳能、施乐、兄弟、爱普生
整机	IBM、惠普、DELL、...
操作系统	Windows、OS X、Linux各大发行版 (海外)
数据库	Oracle、DB2、MariaDB、MySQL
中间件	Weblogic、Websphere、Tuxedo、Jboss、Tomcat
信息安全	KnowBe4、Raytheon、FireEye、RSA、Symantec
应用软件	Autodesk、Adobe、Matlab...



数据库信创产业链/企业（部分）



阿里云/淘宝天猫流量洪峰



开发者社区 > 数据库 > 文章 > 正文

双11特刊 | 一文揭秘云数据库RDS如何顺滑应对流量洪峰

Practice-oriented: Examples : 2019年10月 , OceanBase获得 TPC-C测试第一

Sponsor	System	Performance (TpmC)	Price/TpmC	System Availability	Date Submitted	DB Software
	Alibaba Cloud Elastic Compute Service Cluster	60,880,800	6.25 CNY	10/2/2019	10/1/2019	OceanBase v2.2 Enterprise Edition w/Horizontal Scalab
	SPARC SuperCluster with T3-4 Servers	30,249,688	1.01 USD	6/1/2011	12/2/2010	Oracle Database 11g R2 Enterprise Edition w/Partitioning
	IBM Power 780 Server Model 9179-MHB	10,366,254	1.38 USD	10/13/2010	8/17/2010	IBM DB2 9.7
	SPARC T5-8 Server	8,552,523	0.55 USD	9/25/2013	3/26/2013	Oracle 11g Release 2 Enterprise Edition w/Partitioning
	Sun SPARC Enterprise T5440 Server Cluster	7,646,486	2.36 USD	3/19/2010	11/4/2009	Oracle Database 11g Enterprise Edition w/RAC w/Partitioning
	IBM Power 595 Server Model 9119-FHA	6,085,166	2.81 USD	12/10/2008	6/10/2008	IBM DB2 9.5
	Bull Escala PL6460R	6,085,166	2.81 USD	12/15/2008	6/15/2008	IBM DB2 9.5
	Sun Server X2-8	5,055,888	0.89 USD	7/10/2012	6/20/2012	Oracle Database 11g R2 Enterprise Edition w/Partitioning
	Sun Fire X4800 M2 Server	4,803,718	0.98 USD	6/26/2012	1/18/2012	Oracle Database 11g R2 Enterprise Edition
	HP Integrity Superdome-Itanium2/1.6GHz/24MB iL3	4,092,799	2.93 USD	8/6/2007	2/27/2007	Oracle Database 10g R2 Enterprise Edition w/Partitioning

每分钟完成
6088万笔新
订单，100
万笔/秒订单
；
使用6720个
2.5GHz的处
理器，整体
系统价值3.8
亿元人民币

腾讯云数据库 TDSQL 登顶 TPC-C 榜首

- 2023年3月30日，国际事务处理性能委员会（TPC, Transaction Processing Performance Council）发布最新榜单：腾讯云数据库 TDSQL 刷新纪录
 - 8.14 亿 tpmC，每 tpmC 成本 1.27 元人民币，tpmC 和性价比双榜世界第一

Hardware Vendor	System	v tpmC	Price/tpmC	Watts/KtpmC	System Availability	Database	Operating System	TP Monitor
 Tencent Cloud	Tencent Database Dedicated Cluster (1650 Nodes)	814,854,791	1.27 CNY	NR	06/18/23	Tencent TDSQL v10.3 Enterprise Pro Edition with Partitioning and Physical Replication	Tencent tiLinux 2.2	Nginx 1.16.1
 ANT FINANCIAL	Alibaba Cloud Elastic Compute Service Cluster	707,351,007	3.98 CNY	NR	06/08/20	OceanBase v2.2 Enterprise Edition with Partitioning, Horizontal Scalability and Advanced Compression	Alibaba Aliyun Linux 2	Nginx 1.15.8
 ANT FINANCIAL	Alibaba Cloud Elastic Compute Service Cluster	60,880,800	6.25 CNY	NR	10/02/19	OceanBase v2.2 Enterprise Edition with Partitioning, Horizontal Scalability and Advanced Compression	Alibaba Aliyun Linux 2	Nginx 1.15.8
 ORACLE	SPARC SuperCluster with T3-4 Servers	30,249,688	1.01 USD	NR	06/01/11	Oracle Database 11g R2 Enterprise Edition w/RAC w/Partitioning	Oracle Solaris 10 09/10	Oracle Tuxedo CFSR
 IBM	IBM Power 780 Server Model 9179-MHB	10,366,254	1.38 USD	NR	10/13/10	IBM DB2 9.7	IBM AIX Version 6.1	Microsoft COM+

TPC-C 测试：测试DBMS的性能，模拟一个大型在线事务处理（OLTP）环境，多用户并发执行事务；评估DBMS在高负载情况下的性能，包括吞吐量、响应时间和并发性能等

硬件平台：腾讯云大规模集群
性能tpmC：每分钟处理事务数
性价比Price/tpmC：系统的整体价格与 tpmC 的比值

	DB-Engines 数据库排行	墨天轮国产数据库排行
搜索条目	谷歌、必应	百度、必应、谷歌以及微信公众号文章
趋势指数	谷歌趋势	百度指数、360趋势
技术讨论	Stack Overflow、DBA Stack Exchange	CSDN、墨天轮问答、Stack Exchange、Stack Overflow
招聘相关	Indeed、Simply Hired、LinkedIn	51JOB
博文数目	/	微信公众号、CSDN、墨天轮等
文献数目	/	知网
社交网络	推特	/
加分项目	/	Gartner 份额、IDC 份额、Gartner 魔力象限、TPC-C排行、信通院等

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<https://db-engines.com/en/ranking>

<https://www.modb.pro/dbRank>

Rank			DBMS	Database Model	Score		
Sep 2025	Aug 2025	Sep 2024			Sep 2025	Aug 2025	Sep 2024
1.	1.	1.	Oracle	Relational, Multi-model ⓘ	1170.62	-50.08	-115.97
2.	2.	2.	MySQL	Relational, Multi-model ⓘ	891.77	-23.69	-137.72
3.	3.	3.	Microsoft SQL Server	Relational, Multi-model ⓘ	717.32	-36.84	-90.45
4.	4.	4.	PostgreSQL	Relational, Multi-model ⓘ	657.17	-14.08	+12.81
5.	5.	5.	MongoDB 🟡+	Document, Multi-model ⓘ	380.50	-15.08	-29.74
6.	6.	🟢 7.	Snowflake	Relational	190.19	+11.29	+56.47
7.	7.	🔴 6.	Redis	Key-value, Multi-model ⓘ	145.17	-2.02	-4.25
8.	8.	🟢 9.	IBM Db2	Relational, Multi-model ⓘ	124.19	-3.12	+1.14
9.	9.	🟢 14.	Databricks	Multi-model ⓘ	124.06	+8.25	+39.82
10.	10.	🔴 8.	Elasticsearch	Multi-model ⓘ	118.26	+3.99	-10.53
11.	11.	🔴 10.	SQLite	Relational	107.88	-4.72	+4.53
12.	12.	🔴 11.	Apache Cassandra	Wide column, Multi-model ⓘ	106.98	-1.53	+8.04
13.	13.	🟢 15.	MariaDB 🟡+	Relational, Multi-model ⓘ	91.46	-2.13	+8.02
14.	14.	🔴 12.	Microsoft Access	Relational	83.61	-4.15	-10.15
15.	15.	🟢 17.	Amazon DynamoDB	Multi-model ⓘ	80.28	-3.20	+10.22
16.	16.	16.	Microsoft Azure SQL Database	Relational, Multi-model ⓘ	79.18	+3.34	+6.23
17.	17.	🟢 18.	Apache Hive	Relational	76.10	+5.06	+23.02
18.	18.	🔴 13.	Splunk	Search engine	75.77	+6.00	-17.26
19.	19.	19.	Google BigQuery	Relational	66.00	+0.82	+13.33
20.	20.	🟢 21.	Neo4j	Graph	53.78	-0.69	+11.10

2025年9月数据库排名

2025年8月 墨天轮中国数据库流行度排行榜

- Database System structure

 - DB

 - DBMS, DBMS Functions

 - Three-level structure

 - Data independence

- Data Model

- Database Languages

- Database Users

